SCHOOL OF NATURAL APPLIED SCIENCE

Department of Computer Science and Information Technology (CSIT)

PROJECT – NPRJ740

Advanced ICT Diploma – 2026

CivicLink

Municipal Reporting & Community Intelligence Platform

Phase 1 – Systems Analysis Report

Document Name:	CivicLink – Phase 1 Systems Analysis Report
System Name:	CivicLink
Student Name:	NTHOROANE KP - 202328762
Supervisor:	Dr. Tite Tuyikeze
Submission Date:	01 April 2026

Table of Contents

1. Introduction	3
1.1 Background and Context	3
1.2 Business ICT Assessment	3
2. Identification of ICT Problems	4
2.1 Stakeholder Interview	4
2.2 Interview Questionnaire	5
2.3 Daily Transactions Analysis	6
2.4 System/Software Requirements Analysis	7
2.5 Use Case Diagram	8
2.6 Solution Analysis	9
3. Conclusion	11

1. Introduction

1.1 Background and Context

Sol Plaatje Local Municipality, headquartered in Kimberley in the Northern Cape Province of South Africa, is responsible for delivering essential services to a population of approximately 250,000 residents. The municipality manages water distribution, electricity supply, road maintenance, waste collection, and public safety coordination across multiple residential, commercial, and industrial zones.

Kimberley, as the provincial capital of the Northern Cape, experiences a unique set of service delivery challenges stemming from aging infrastructure, rapid urbanisation in informal settlements, and a historically under-resourced maintenance workforce. The city regularly faces acute incidents including burst water pipes, prolonged electricity outages, illegal dumping, and increasingly — public protests and worker strikes that disrupt road access and municipal operations.

On the date of this report (31 March 2026), a municipal workers' strike resulted in major road blockages across key arterial routes in Kimberley, including routes used by taxis and commuters. Residents were forced to seek alternative routes with no official guidance, highlighting the absence of a real-time public communication system within the municipality.

1.2 Business ICT Assessment

Sol Plaatje Municipality currently relies on a combination of telephonic complaint lines, walk-in service centres, and informal social media channels (particularly WhatsApp and Facebook groups) for residents to report service delivery issues. This fragmented approach results in:

- Delayed response times due to manual triaging of complaints
- Duplicate report submissions with no deduplication logic
- No GPS-based routing of incidents to the correct department
- Zero real-time notification capability for residents tracking their reports
- No centralised visibility for municipal management to prioritise resources
- No mechanism for broadcasting urgent alerts (e.g., road closures, strike updates) to affected residents

The municipality has no integrated digital platform that connects residents, field workers, and administrative officials in a single real-time system. This gap directly contributes to service delivery failures, public frustration, and the escalation of minor infrastructure issues into major crises.

2. Identification of ICT Problems

2.1 Stakeholder Interview

Interview Summary

A mock interview was conducted with a senior official at Sol Plaatje Municipality to understand the current service delivery reporting process and identify systemic ICT gaps. The interview targeted the Infrastructure and Community Services division, which oversees water, electricity, roads, and waste management.

Interviewee	Senior Infrastructure Official – Sol Plaatje Municipality
Date	31 March 2026
Location	Kimberley, Northern Cape
Purpose	Identify ICT gaps in service delivery reporting systems

Key Findings from Interview

The following critical problems were identified through the interview:

- Water infrastructure: Burst pipes and water leakages are reported daily, but no digital tracking system exists. Calls are logged manually and often lost or misdirected between departments.
- Electricity outages: Outages affecting large areas of Kimberley are communicated via word of mouth. The municipality has no push notification system to alert residents of planned or unplanned outages.
- Illegal dumping: Hotspot areas for illegal dumping are known but not formally mapped. Field teams are often dispatched to the wrong locations due to imprecise verbal descriptions.
- Road blockages and strikes: On 31 March 2026, a municipal workers' strike caused multiple road closures. No official system existed to notify commuters or suggest alternative routes, leading to widespread confusion.
- Duplicate reporting: Residents report the same burst pipe or outage multiple times across different channels, wasting call centre resources and distorting the priority queue.
- No feedback loop: Residents receive no updates on the status of their reports, causing mistrust and repeat calls that overwhelm the system.

2.2 Interview Questionnaire

The following structured questionnaire was used during the stakeholder interview:

#	Question	Response Summary
1	How do residents currently report service delivery issues?	Primarily via phone to the call centre or walk-in; some via social media groups.
2	What are the most frequently reported infrastructure problems?	Burst pipes, water leakages, electricity outages, and potholes are the top four.
3	How does the municipality track and assign reported issues?	Manually via spreadsheets and phone calls. No centralised digital system.
4	Are residents notified when their report is resolved?	No formal notification system. Residents must call back to check status.
5	How does the municipality communicate urgent alerts (strikes, outages)?	Social media posts and radio announcements — no targeted or geo-fenced notifications.
6	What happened today with the strike-related road closures?	Major roads blocked, taxis rerouted. No official digital communication to commuters.
7	Are illegal dumping sites tracked or mapped?	Not formally. Field teams rely on verbal reports with no GPS coordinates.
8	What is the average time to respond to a reported burst pipe?	Between 24–72 hours, often longer due to poor routing to the correct team.
9	Do field workers receive digital task assignments?	No. Supervisors call field workers directly. No tracking of task completion.
10	What digital solution would most improve your operations?	A single platform where residents report, managers assign, and workers update status in real time.

2.3 Daily Transactions Analysis

Monitoring of Sol Plaatje Municipality's daily operational transactions reveals a consistent pattern of inefficiencies rooted in the absence of a unified digital reporting infrastructure. The following transactions occur daily:

Transaction Type	Current Process	ICT Problem Identified
Resident Report Submission	Phone calls or walk-ins logged manually by call centre agents.	No digital capture, GPS, or media upload. High error rate.
Burst Pipe Reporting	Resident calls call centre; agent writes on paper/spreadsheet.	Duplicates, lost entries, no location accuracy.
Electricity Outage Management	Outages logged per call; no aggregation or alerting.	No push alerts, no affected area mapping.
Illegal Dumping Report	Verbal description of location; supervisor dispatches team.	Inaccurate location data; repeat visits to same site.
Task Assignment to Field Workers	Supervisor phones field worker directly.	No audit trail, no status updates, no accountability.
Strike / Protest Alert (31 Mar 2026)	Radio announcement and WhatsApp groups only.	No geo-targeted alerts; commuters unaware of alternative routes.
Road Closure Communication	Physical signage placed by workers on site.	No real-time digital map update; travellers not rerouted.
Report Status Update to Resident	Resident calls back to enquire; no proactive feedback.	No status tracking, no notification system.

These transactions reveal a municipality operating entirely on reactive, analogue processes. The lack of ICT integration results in lost data, inflated response times, and a breakdown of trust between the municipality and the communities it serves.

2.4 System / Software Requirements Analysis

Based on the interview findings and daily transaction analysis, the following functional and non-functional requirements have been identified for a proposed digital solution:

Functional Requirements

- Residents must be able to submit incident reports with GPS location, category, description, and photo/video evidence.
- The system must support incident categories: Water (burst pipe, leakage), Electricity (outage), Roads (pothole, blockage), Waste (illegal dumping), and Public Safety (strike, suspicious activity).
- Municipality officials must be able to view, triage, and assign reports to relevant departments.
- Field workers must receive assigned tasks digitally and update job status in real time.
- The system must push geo-targeted notifications to residents in affected areas.
- Residents must be able to track the status of their submitted reports (Submitted → In Progress → Resolved).
- The system must support emergency alert broadcasting by municipality officials.
- An analytics dashboard must provide heatmaps, resolution rates, and response time metrics.

Non-Functional Requirements

- Availability: The system must be accessible 24/7 with at least 99.5% uptime.
- Performance: Reports must be stored and visible to admins within 5 seconds of submission.
- Security: Role-based access control (Resident, Official, Field Worker, Admin). All data encrypted in transit and at rest.
- Scalability: The system must support at minimum 10,000 concurrent users across the Sol Plaatje municipal area.
- Usability: The resident interface must be accessible on low-end Android devices with minimal data usage.
- Offline capability: Basic report drafting must function offline, with sync when connectivity is restored.

Technology Justification

A Firebase-based architecture is proposed due to its real-time database synchronisation, built-in authentication, scalable cloud functions, and native push notification support via Firebase Cloud Messaging (FCM). These capabilities directly address the identified ICT problems without requiring significant on-premises infrastructure investment from the municipality.

2.5 Use Case Diagram

The following Use Case Diagram illustrates the interactions between the four primary system actors and the CivicLink platform's core use cases. Three actors are identified: the Resident who submits reports and receives alerts; the Municipality Official who manages reports, assigns tasks, and broadcasts alerts; and the Field Worker who receives and updates assigned tasks.

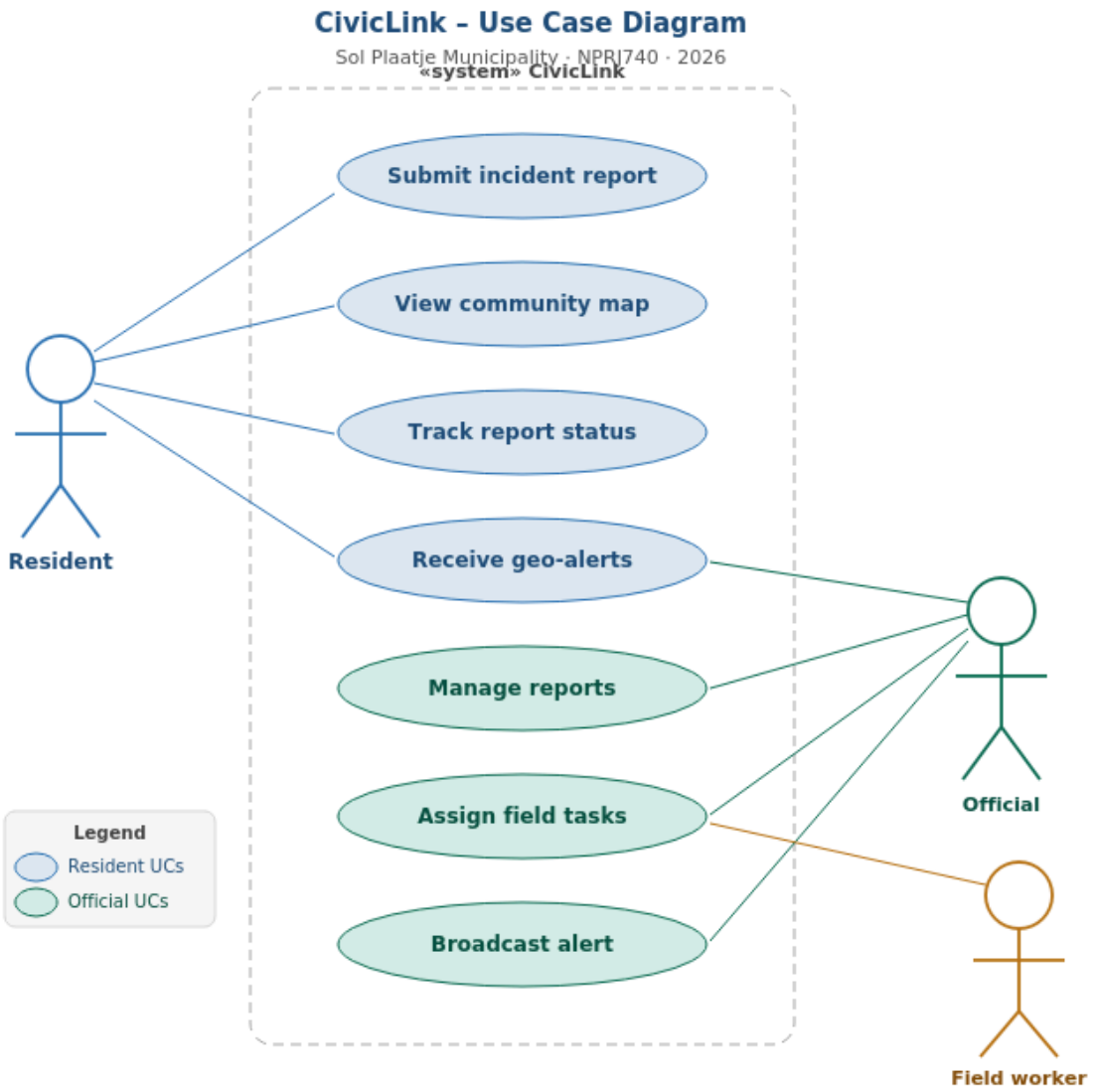


Figure 1: CivicLink UML Use Case Diagram – Sol Plaatje Municipality

2.6 Solution Analysis

Based on the identified ICT problems, the following solutions are proposed and evaluated:

#	Problem	Proposed Solution	Technology
1	No digital report submission	PWA mobile app allowing GPS-tagged incident reports with photo upload.	React/Next.js PWA, Firebase Storage, Google Maps API
2	No real-time alert system	Geo-targeted push notification broadcasts for outages, strikes, and road closures.	Firebase Cloud Messaging (FCM), Firestore geo-queries
3	No issue tracking or feedback	Resident dashboard with live status updates (Submitted → In Progress → Resolved).	Firestore real-time listeners, Cloud Functions
4	Inaccurate location reporting (illegal dumping)	GPS coordinates captured automatically on report submission; displayed on live map.	Google Maps API, Geolocation API
5	No task management for field workers	Admin assigns tasks digitally; field worker updates status via mobile interface.	Firestore, Cloud Functions, Firebase Auth (role-based)
6	No management analytics	Admin dashboard with heatmaps, KPIs, response times, and issue frequency charts.	React + Recharts, Firestore aggregation queries

Recommended Solution: CivicLink Platform

The recommended solution is CivicLink — a fullstack, Firebase-based Progressive Web Application that unifies all six solution components into a single, cohesive platform. CivicLink is designed to serve the Sol Plaatje Municipality and is scalable to other South African municipalities facing similar service delivery challenges.

The diagram below illustrates how an incident report flows through the CivicLink system from submission to resolution, as well as how the municipality broadcasts emergency alerts — such as the 31 March 2026 strike — to affected residents in real time:

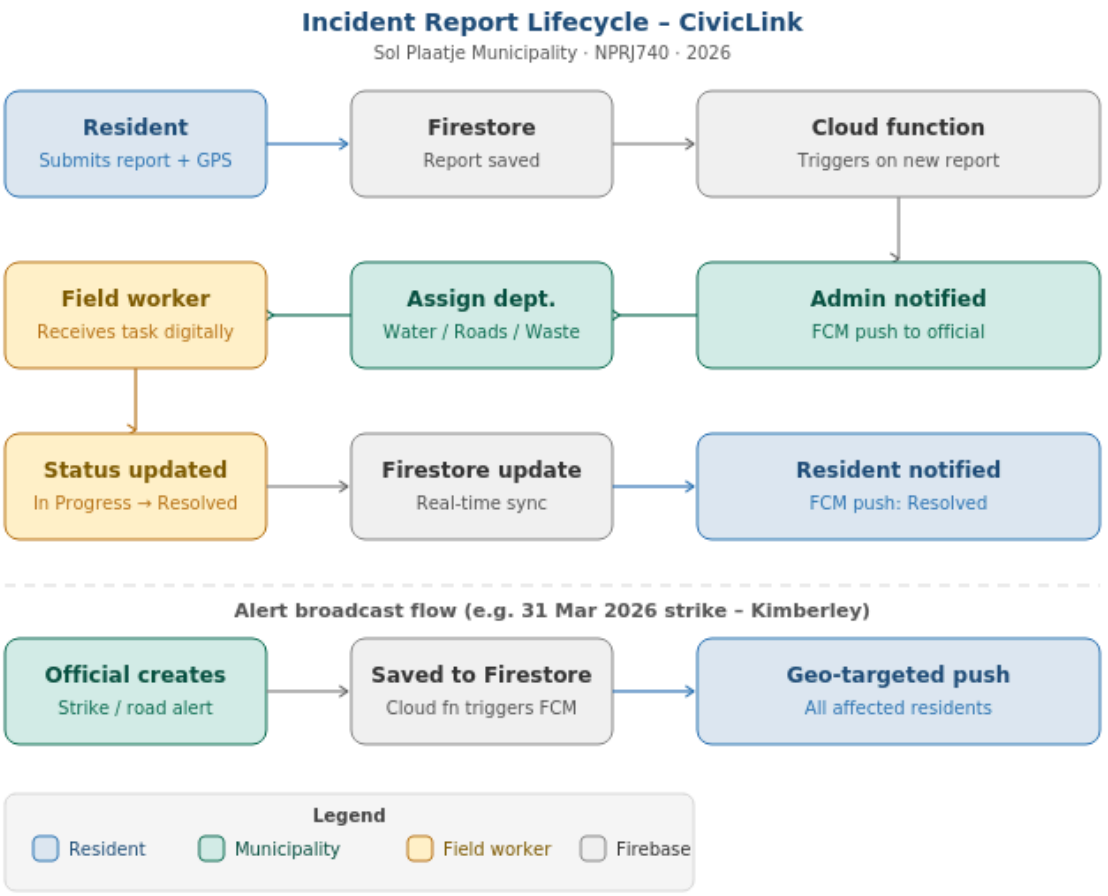


Figure 2: CivicLink Incident Report Lifecycle & Alert Broadcast Flow

CivicLink's architecture consists of: (1) a Resident PWA for report submission and alert reception, (2) a Municipality Admin Portal for incident management and analytics, and (3) a Field Worker interface for task management. All components communicate via Firebase's real-time infrastructure, ensuring immediate data synchronisation across all user roles.

3. Conclusion

Sol Plaatje Municipality in Kimberley currently operates its service delivery reporting function through manual, fragmented, and analogue processes. The investigation — conducted through stakeholder interviews and daily transaction analysis — has revealed fundamental ICT deficiencies that directly impede the municipality's ability to respond to community needs efficiently.

The most critical problems identified are: the absence of a digital incident reporting system with GPS and media capture capabilities; the lack of real-time push notification infrastructure for broadcasting urgent alerts to residents; no digital task assignment and tracking system for field workers; and no feedback mechanism for residents to track the resolution of their reported issues. These failures were starkly demonstrated on 31 March 2026, when a municipal workers' strike caused widespread road blockages in Kimberley, and the municipality had no digital means to alert commuters or suggest alternative routes in real time.

The proposed solution — CivicLink — directly addresses each identified problem through a Firebase-based fullstack platform accessible via web and mobile. By enabling residents to report incidents with GPS precision, providing municipality officials with real-time dashboards for triage and task assignment, enabling geo-targeted alert broadcasts, and closing the feedback loop with residents through automated status notifications, CivicLink transforms the municipality's operational model from reactive to proactive.

The implementation of CivicLink will not only improve internal municipal efficiency and accountability, but will restore public trust in service delivery by making the municipality's actions visible, traceable, and responsive to community needs. Phase 2 of this project will detail the system design specifications, Firestore schema, component architecture, and development roadmap required to bring CivicLink to production.

— End of Phase 1 Report —